

that the most critical application enabled by programmability of GPUs is image and video stabilisation. He adds, "Digital still and video cameras are common gadgets and not all of the users are professionals. The images are blurred and shaky, especially when using zoom for far away objects. So a lot of software algorithm supported by GPUs is loaded on these gadgets for image and video stabilisation. Techniques like processing each frame and removing the unwanted ones, and matching brightness and colour are used to give an enjoyable experience to the user."

Facial tagging. Digitisation has added not only lots of features to photography but also a lot more photographs to our albums. Over the years, when you have 50,000-60,000 pictures, it is very difficult to find all the pictures of a particular person if you had not tagged them initially.

"Thanks to highly intelligent and smart facial tagging software algorithms, you can tag all the 50,000 pictures with just one click," says Krishna. These are intensive computing applications which need a capable graphics chip as well as good software.

Video transcoding. Video transcoding allows you to watch an HD video on your HD TV set as well as on your Android-based smartphone with ten times smaller screen. Basically, the algorithm allows you to convert a 4GB file into a 100MB file when you view the same video on different types of screens.

Krishna informs, "A quad-core CPU will do the same task in 2-2.5 hours, but with massively parallel GPUs, the conversion is done within 15-20 minutes seamlessly."

Video upscaling. It is another algorithm very critical in plasma and LCD TVs available today. The DVD collec-

tion of your video library is probably suitable for standard definition (SD), but you cannot replace every SD-DVD with an HD video to view on your new plasma TV.

"A lot of algorithm goes in your HD TV set to support your old SD videos. It basically uses interpolation and adds a lot of data to existing data so that you get near-high-definition experience."

Gesture recognition. Today, there are a number of sensors added to your phone, laptop and desktop PC in order to perform various tasks like gesture recognition, motion recognition, eye tracking and multi-gesture recognition.

Bhandari shares, "These kinds of applications use three 3D cameras and all the data is captured and processed in real time. Even playing a fast-paced car game requires a millisecond response time. Such applications need a good balance of computing and graphics both."

Every improvisation on a video basically involves parallel processing capabilities. And GPUs today offer intense hardware capability fit for these applications. It's now the software and algorithm that need to exploit the hardware capability to provide the best visual experience.

Software: Key to unlock hardware potential

The software is an important element that helps determine the performance and efficiency of a given product. With the advent of DirectX and OpenGL (open graphics library), GPUs can process short programs for each pixel, adding programmability to the chip.

"Millions of applications that are already there in the market will continue to drive the classic microprocessor/microcontroller industry. Where

massively-parallel applications start choking the CPU capabilities, the software will take the advantage of GPUs. This is where open computing language (OpenCL) comes into picture. OpenCL is a framework for writing programs that execute across heterogeneous platforms consisting of CPUs, GPUs and other processors," explains Krishna.

For optimal utilisation of the GPU, the software design plays a crucial role. The same GPU chip can perform very differently when it comes to using software. Justifying this, Agrawal says, "A high-end Nvidia GPU can perform worse than an Intel CPU like i3 or i5 if the software designer doesn't know how to use the parallel processing being provided by CUDA in Nvidia GPU, or he is unable to break the software in parallel processing units. It also matters what kind of software you are designing. A software which has no inherent parallel blocks can perform better on a GPU needing non-parallel scheduling like Intel's."

GPUs for rich experience

"India is a country which prizes top-class entertainment very highly. Expectedly, increasingly rich, technologically advanced entertainment is becoming a major driver for growth in the graphics industry," says Goyal.

Whether it is movies and TVs moving from standard- to high-definition, games shifting from 2D to stereoscopic 3D, or the explosion of demanding Flash-based Web applications like Facebook and games such as Farmville, all these trends require the GPU. Without GPUs' parallelism, it is impossible to develop power-efficient supercomputers, experts believe. ●

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